UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 08/06/2023*

We observe two samples of a realization of the random process $X[n]=S[n]+W[n]$, for n=0 and n=1. $S[n]=A$ is the signal of interest (SOI), A is modelled as a unknown deterministic parameter, $W[n]$ is a w.s.s. zero-mean Gaussian process modelled as an Auto-Regressive (AR) process of order 1, i.e. $W[n]=\rho W[n-1]+I[n]$, where ρ is the one-lag correlation coefficient of the noise process $W[n]$ and $I[n]$ is the white innovation process with power σ_I^2.

(1) Derive the mean η_X, the Autocorrelation Function (ACF) $R_X[m]$, and the power spectral density (PSD) $S_X\left(e^{j2\pi f}\right)$ of the process $X[n]$ and plot them.

(2) Derive the Probability Density Function (PDF) of the data vector $\mathbf{X}=\left[X[0] \quad X[1]\right]^T$, the log-likelihood function, $\ln LF(a)$, and the Maximum Likelihood (ML) estimator of the parameter *A*, given the observation of $\mathbf{X}$.

(3) Derive the bias and the Mean Square Error (MSE) of the ML estimator $\hat{A}_{ML}$. Plot the MSE as a function of the one-lag correlation coefficient ρ and explain what happens to $\hat{A}_{ML}$ and its MSE when $\rho=-1$, $\rho=0$, and $\rho=1$.

(4) Derive the Cramér-Rao lower bound (CRB) on the estimate of *A* and establish whether $\hat{A}_{ML}$ is efficient or not.

Solution

(1) Mean, ACF and PSD of the process X[n]:

$$\eta_X = E\{X[n]\} = E\{A + W[n]\} = E\{A\} + E\{W[n]\} = A$$

$$R_X[m] = E\{X[n]X[n+m]\} = E\{(A + W[n])(A + W[n+m])\}$$
$$= E\{A^2\} + E\{W[n]W[n+m]\} = A^2 + \sigma_W^2 \rho^{|m|}$$

$$S_X\left(e^{j2\pi f}\right) = FT\left\{A^2 + \sigma_W^2 \rho^{|m|}\right\} = FT\left\{A^2\right\} + FT\left\{\sigma_W^2 \rho^{|m|}\right\} = 2\pi A^2 \delta\left(e^{j2\pi f} - 1\right) + \frac{\sigma_W^2 (1-\rho^2)}{1+\rho^2 - 2\rho\cos(2\pi f)}$$

In fact, the ACF and PSD of the noise process W[n] can be derived as follows:

$$W[n] = \rho W[n-1] + I[n]$$

$$W(z) = \rho z^{-1} W(z) + I(z) \quad \Rightarrow \quad L(z) = \frac{W(z)}{I(z)} = \frac{1}{1-\rho z^{-1}} = \frac{z}{z-\rho}$$

The innovation filter has 1 zero at $z = 0$ and 1 pole at $z = \rho$. Hence, the innovation filter is a high-pass filter if $\rho < 0$ and a low-pass filter if $\rho > 0$, and so the process.

The PSD and the ACF are given by (see the course vu-graphs for the derivation):

$$S_W\left(e^{j2\pi f}\right) = \sigma_I^2 \left|L\left(e^{j2\pi f}\right)\right|^2 = \sigma_I^2 \left|\frac{1}{1-\rho e^{-j2\pi f}}\right|^2 = \frac{\sigma_I^2}{1+\rho^2 - 2\rho\cos(2\pi f)} = \frac{\sigma_W^2 (1-\rho^2)}{1+\rho^2 - 2\rho\cos(2\pi f)}$$

$$R_W[m] = FT^{-1}\left\{S_W\left(e^{j2\pi f}\right)\right\} = \frac{\sigma_I^2}{1-\rho^2}\rho^{|m|} = \sigma_W^2 \rho^{|m|}, \quad \text{where } \sigma_W^2 = R_W[0] = \frac{\sigma_I^2}{1-\rho^2}.$$

As previously stated, when $\rho < 0$, $W[n]$ is a high-pass process, when $\rho > 0$, $W[n]$ is low-pass.

(2) PDF of X and ML estimator:

X is a 2-D Gaussian random vector: $\mathbf{X} \in N\left(A\mathbf{1}, \sigma_W^2 \mathbf{M}\right)$ where the normalized covariance matrix **M** is given by: $\mathbf{M} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$. Its inverse $\mathbf{M}^{-1}$, which will be used later, is easily derived:

$$\mathbf{M}^{-1} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}^{-1} = \frac{1}{1-\rho^2}\begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}.$$

The pdf in vector format is:

$$f_{\mathbf{X}}(\mathbf{x};A)=\frac{1}{\sqrt{(2\pi)^2\left|\sigma_W^2\mathbf{M}\right|}}\exp\left(-\frac{1}{2}(\mathbf{x}-A\mathbf{1})^T\left(\sigma_W^2\mathbf{M}\right)^{-1}(\mathbf{x}-A\mathbf{1})\right)$$
$$=\frac{1}{\sqrt{\left(2\pi\sigma_W^2\right)^2|\mathbf{M}|}}\exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x}-A\mathbf{1})^T\mathbf{M}^{-1}(\mathbf{x}-A\mathbf{1})\right)$$

where $|\mathbf{M}|=1-\rho^2$ and $\mathbf{M}^{-1}=\frac{1}{1-\rho^2}\begin{bmatrix}1 & -\rho\\ -\rho & 1\end{bmatrix}$.

The ML estimate is obtained as the solution of the ML equation:

$$\ln LF(a)=\ln f_{\mathbf{X}}(\mathbf{x};a)=\kappa_1-\frac{1}{2\sigma_W^2}(\mathbf{x}-a\mathbf{1})^T\mathbf{M}^{-1}(\mathbf{x}-a\mathbf{1})$$
$$=\kappa_1-\frac{1}{2\sigma_W^2}\mathbf{x}^T\mathbf{M}^{-1}\mathbf{x}-\frac{a^2}{2\sigma_W^2}\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}+\frac{1}{2\sigma_W^2}a\mathbf{1}^T\mathbf{M}^{-1}\mathbf{x}+\frac{1}{2\sigma_W^2}a\mathbf{x}^T\mathbf{M}^{-1}\mathbf{1}$$
$$=\kappa_2-\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}}{2\sigma_W^2}a^2+\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{x}}{\sigma_W^2}a$$

$$\frac{d}{da}\left(\ln f_{\mathbf{X}}(\mathbf{x};a)\right)=-\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}}{\sigma_W^2}a+\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{x}}{\sigma_W^2}=0$$

$$\hat{A}_{ML}=\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{x}}{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}}=\mathbf{h}^T\mathbf{x},\quad \text{where } \mathbf{h}\triangleq\frac{\mathbf{M}^{-1}\mathbf{1}}{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}}=\frac{\begin{bmatrix}1 & -\rho\\ -\rho & 1\end{bmatrix}\mathbf{1}}{\mathbf{1}^T\begin{bmatrix}1 & -\rho\\ -\rho & 1\end{bmatrix}\mathbf{1}}=\frac{1}{2-2\rho}\begin{bmatrix}1-\rho\\ 1-\rho\end{bmatrix}=\begin{bmatrix}\frac{1}{2}\\ \frac{1}{2}\end{bmatrix}=\frac{1}{2}\mathbf{1}$$

Hence, $\hat{A}_{ML}=\mathbf{h}^T\mathbf{x}=\frac{1}{2}\mathbf{1}^T\mathbf{x}=\frac{X[0]+X[1]}{2}$.

Note that the structure of the ML estimator does not depend on the value of ρ. Moreover, $\hat{A}_{ML}$ is a **linear** function of the data vector **X**, it is the well-known Whitening Matched Filter (WMF). Since the estimate is a linear function of the Gaussian distributed data vector **X**, the estimate is Gaussian distributed, and consequently the estimation error is Gaussian distributed, with mean and variance equal to the bias and the MSE, respectively.

(3) Bias and MSE of the ML estimator of *A*:

$E\left\{\hat{A}_{ML}\right\}=E\left\{\frac{X[0]+X[1]}{2}\right\}=\frac{A+A}{2}=A$. Hence, the estimator is unbiased.

$$MSE\left\{\hat{A}_{ML}\right\} = E\left\{\left(A-\hat{A}_{ML}\right)^2\right\} = E\left\{\left(A-\frac{X[0]+X[1]}{2}\right)^2\right\} = E\left\{\left(A-\frac{A+W[0]+A+W[1]}{2}\right)^2\right\}$$

$$= E\left\{\left(-\frac{W[0]+W[1]}{2}\right)^2\right\} = \frac{1}{4}E\left\{W^2[0]+W^2[1]+2W[0]W[1]\right\}$$

$$= \frac{1}{4}\left(\sigma_W^2+\sigma_W^2+2\rho\sigma_W^2\right) = \frac{1+\rho}{2}\sigma_W^2$$

$$MSE\left\{\hat{A}_{ML}\right\}\Big|_{\rho=-1} = 0, \quad MSE\left\{\hat{A}_{ML}\right\}\Big|_{\rho=0} = \frac{\sigma_W^2}{2}, \quad MSE\left\{\hat{A}_{ML}\right\}\Big|_{\rho=1} = \sigma_W^2$$

(5) CRB on the estimate of *A*:

$$\frac{d^2 \ln f_{\mathbf{X}}(\mathbf{x};a)}{da^2} = \frac{d}{da}\left[\frac{d}{da}\left(\ln f_{\mathbf{X}}(\mathbf{x};a)\right)\right] = \frac{d}{da}\left[-\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}}{\sigma_W^2}a+\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{x}}{\sigma_W^2}\right] = -\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}}{\sigma_W^2}$$

$$CRB(A) = \frac{1}{-E\left\{\dfrac{d^2 \ln f_{\mathbf{X}}(\mathbf{x};a)}{da^2}\right\}} = \left(\frac{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}}{\sigma_W^2}\right)^{-1} = \frac{\sigma_W^2}{\mathbf{1}^T\mathbf{M}^{-1}\mathbf{1}} = \frac{1-\rho^2}{2-2\rho}\sigma_W^2 = \frac{1+\rho}{2}\sigma_W^2 = MSE\left\{\hat{A}_{ML}\right\}$$

Hence, the estimator is **efficent**.